Bacon Networking Configuration

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This report discusses the networking solutions for the Bacon brand to support 75MBs data transfer. It specifies the improvements that need to be made in Orlando, Toronto, and Phoenix where each floor covers an area of 250 by 400ft. Each site will have a total of 150 VLAN networks connected to two subnets, which will support management and security. The admin will also have a unique subnet to control access located on the 3rd floor. This is to host different departments such as IT, Human Resource, Management, Legal affairs among many others. Each department will be registered and assigned a specific VLAN ID. Three subnets will be implemented where each subnet will host multiple networks. To support internetwork communication, VLANs will be effective because it would build a virtual net by itself. Once all this put in place, a firewall will be implemented on each site between the router and the layer 3 switch. Though the implementation of the firewall gives good security, encryption of all passwords and confidential data, implementation of port security on the switchports and consideration of proper security management of any unused ports in all of the switches and other end user interfaces provide more security to the Bacon network. To improve the network issues at Bacon, possible solutions are discussed alongside the infrastructure required.

Orlando

Current technological issues in Orlando

Since Orlando is the main data center, the network will be integrated to have 150 network connections in the first and second floors of the building. The third floor would require a 75 number of networks. The system will be divided by enabling VLAN protocols to the port interfaces that would support all the networks (Degadzor et al, 2015). The VLAN configured will be protocol based on the order to handle traffic effectively. That is, the networks will be controlled based on the traffic protocol. The purpose of having multiple networks is to share resources such as computers, printers and scanners.

Proposed solutions

The Orlando contains CIDR/21 with 2046 of usable hosts. This will contain the three main VLANs including the internet VLAN, Production VLAN and the Management VLAN. Some of the important functions of this site will be to host Web applications and Database servers. To support transfer services, the bandwidth of the network will be 75Mbps.

Orlando VLAN

VLAN	**Usable Range**	**Host**	**Subnet Mask**	**CIDR Value**	**Broadcast**	**Net ID**
Management10	192.168.1.1 - 192.168.1.254	150	255.255.255.0	255.255.255.0/24	192.168.1.255	192.168.1.0
Internet 20	192.168.99.1 192.168.99.254	150	255.255.255.0	255.255.255.0/24	192.168.99.255	192.168.99.0
Production 30	192.168.11.1 - 192.168.11.126	75	255.255.255.128	255.255.255.128/25	192.168.11.127	192.168.11.0

Phoenix

This site will be to accommodate up to 4094 users which will be facilitated by the CIDR/20. Like Orlando, the site will also contain internet VLAN, Production VLAN and Management VLAN. This site will contain six switches, two firewalls and two routers. Considering that Phoenix is divided into two buildings, one part will specifically host the failover network for the system (Smailes, 2017). There will also be a backup data center that will have the additional network configured. The extra network will have a limited IP address for the private connection.

Phoenix 1 VLAN

VLAN	Usable Range	Host	Subnet Mask	CIDR Value	Broadcast	Net ID
Manageme nt40	192.168.2.1 - 192.168.2.254	150	255.255.255.0	255.255.255.0/24	192.168.2.255	192.168.2.0
Internet 50	192.168.19.1 192.168.19.254	150	255.255.255.0	255.255.255.0/24	192.168.19.255	192.168.19.0
Production 60	192.168.22.1 - 192.168.22.126	75	255.255.255.128	255.255.255.128/ 25	192.168.22.127	192.168.22.0

Phoenix 2 VLAN

VLAN	Usable Range	Host	Subnet Mask	CIDR Value	Broadcast	Net ID
Manageme nt10	192.168.3.1 - 192.168.3.254	150	255.255.255.0	255.255.255.0/24	192.168.3.255	192.168.3.0
Internet 20	192.168.77.1 192.168.77.254	150	255.255.255.0	255.255.255.0/24	192.168.77.255	192.168.77.0
Production 30	192.168.33.1 - 192.168.33.126	75	255.255.255.128	255.255.255.128/ 25	192.168.33.127	192.168.33.0

Toronto

Like Orlando, this site will accommodate a total of 2,046 total usable hosts. There will be three VLANs like other sites. This site will be used to as a database of production activities. In this site, three switches, a firewall and a router will be used to construct the network. In addition, the system will be implemented using TCP/IP protocol in order to support the use of VLANs, which applies in the data link layer. To speed up the operations and enhance effective networking, the Open Shortest Path First protocol will be applied to enable efficient routing of data (Haresh & Prof. Rashmi, 2014).

VLAN	Usable Range	Host	Subnet Mask	CIDR Value	Broadcast	Net ID
Management10	192.168.4.1 - 192.168.4.254	150	255.255.255.0	255.255.255.0/24	192.168.4.255	192.168.4.0
Internet 20	192.168.66.1 192.168.66.254	150	255.255.255.0	255.255.255.0/24	192.168.66.255	192.168.66.0
Production 30	192.168.44.1 - 192.168.44.126	75	255.255.255.128	255.255.255.128/25	192.168.44.127	192.168.44.0

Meanwhile, other technologies will be implemented for efficient network connectivity. These include EIGRP for assisted routing of data through an efficient path, RIP will also assist in updating status of routing through assisted communication with other routers in the network. The OSFP will be used in rerouting (Manzoor, Hussain & Mehrban, 2020).

Conclusion

In conclusion, Bacon will be improved by incorporating VLAN in all three locations. Orlando and Phoenix will have private networks because they host the data center and failover data center. This way, the servers can be easily switched in case of failure. In sum, the Bacon networking system have been improved by improvising it to host 150 networks.

References

Degadzor F., A., Adamu M., M., Collinson C. M., A., Botchey F., E., &Anibrika S.K. Bright Prof.Somuah, C. (2015). Performance analysis of interior gateway routing protocol eigrpover open shortest path first OSPF protocol. *International Journal of scientific & technology research (IJSTR)*, *5*(9), 111-117. Doi: 10.14569/IJACSA.2016.070512

Haresh N., P., & Prof. Rashmi, P. (2014). Extensive reviews of OSPF and EIGRP routing protocols based on route summarization and route redistribution. *International Journal of Engineering Research and Applications*, *4*(9), 141-144. Doi: 10.15680/IJIRCCE.2015

Manzoor, A., Hussain, M., & Mehrban, S. (2020). Performance Analysis and Route Optimization: Redistribution between EIGRP, OSPF & BGP Routing Protocols. *Computer Standards & Interfaces*, 68. https://doi-org.ezproxy.umuc.edu/10.1016/j.csi.2019.103391

Smailes, M., Chong, N., Mckeever, P., Shek, J., Theotokatos, G., &Abusara, M. (2017). *Hybrid, Multi-Megawatt HVDC Transformer Topology Comparison for Future Offshore Wind Farms*.